L4a: Lattice Trading Rules and Probability of Profit

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we turn a lattice forecast into a precisely defined terminal trading event, compute its probability, and extend the lattice beyond two branches.

Today's concepts

- Specify a terminal rule: identify the dated cash flows, holding period, benchmark growth rate, costs, and target.
- Derive a target probability: convert a strict return target into a minimum up-move count and evaluate a binomial tail.
- Generalize the state space: distinguish terminal and first-passage exits, then represent an N-ary lattice with branch counts.

Lectures and Examples

Lecture:
CHEME-5660-L4a-Lecture-LatticeModel-TradeRule-Fall-2026.ipynb

Example 1:
CHEME-5660-L4a-Example-CumulativeProbabilityLattice-Fall-2026.ipynb

Example 2:
CHEME-5660-L4a-Example-N-Aray-Lattice-Fall-2026.ipynb

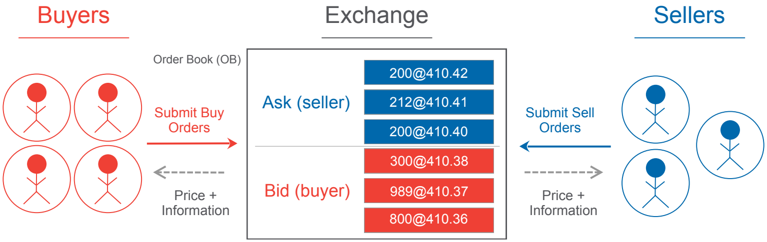
The first example estimates a binomial lattice and evaluates a strict terminal target. The second discretizes empirical growth into three branches and plots the probability mass at a selected lattice level.

Concept Review: Exchanges, Stocks, and Orders

- A stock is an ownership share in a company.
- An exchange is a regulated marketplace for securities.
- An ETF is a basket of assets whose shares trade on an exchange.
- A mutual fund pools investor capital and is priced periodically, usually once each trading day.
- A market order seeks immediate execution, but its price is not guaranteed.
- A limit order specifies a price or better, but execution is not guaranteed.
- A market maker posts bid and ask prices, helping other participants trade.

Execution matters: the bid–ask spread and order type affect the cash flows in a complete trading rule.

The Exchange Connects Market Participants



Link: live CBOE order book viewer

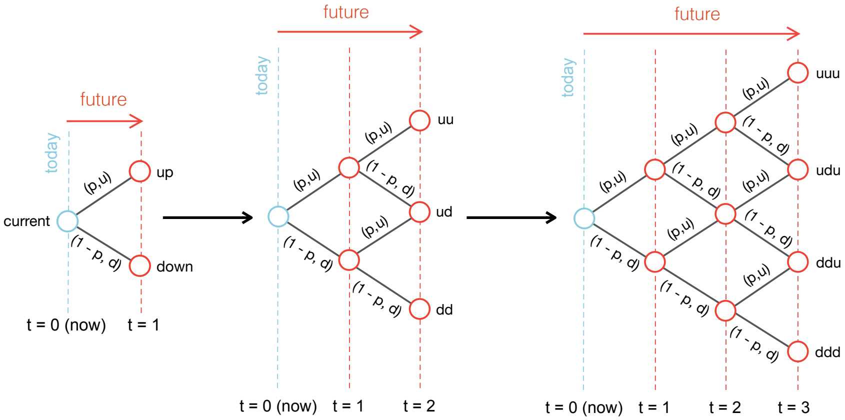
Company Profile: Citadel Securities

Citadel Securities is a market maker. A market maker posts prices at which it is willing to buy and sell, allowing other participants to trade.

- The bid is the price at which the market maker will buy.
- The ask is the price at which the market maker will sell.
- The ask is normally above the bid. Their difference is the bid–ask spread.
- A complete trade model uses executable entry and exit prices, not one abstract price for both sides of the transaction.

Role in this lecture: the market setting motivates the cash flows and costs that must be specified before computing probability of profit.

A Recombining Price Lattice



1-step ahead prediction

2-step ahead prediction

3-step ahead prediction

The Binomial Lattice Model

Let $S_0 > 0$ be today's share price. At each step, multiply the price by a constant up factor $u > 0$ with probability p, or a constant down factor $d > 0$ with probability $1 - p$, where $u > d$.

After t steps, let K_t count the up moves. Then the node indexed by $K_t = k$ has price and real-world probability:

$$\underbrace{S_{t,k} = S_0 u^k d^{t-k}}_{\text{node price}}, \qquad \underbrace{\mathbb{P}(K_t = k) = \binom{t}{k} p^k (1-p)^{t-k}}_{\text{node probability}}.$$

The model is transparent, but fixed u, d, and p do not reproduce heavy tails or volatility clustering.

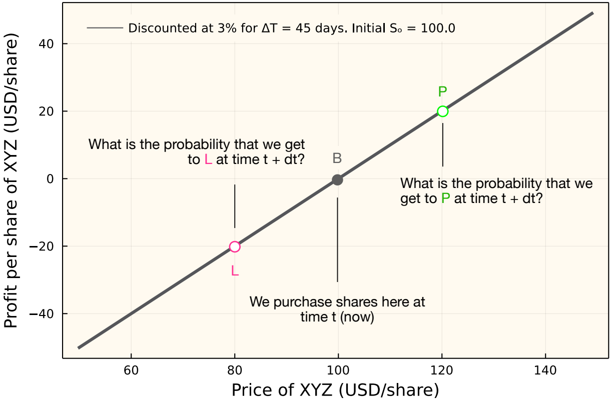
A Forecast Is Not Yet a Trading Rule

A price model describes possible states and their probabilities. A trading rule must add the decisions that define the event of interest.

- When do we enter, and how many shares do we buy?
- Is the exit scheduled at a terminal date or monitored before it?
- Which entry and exit prices represent executable prices?
- What benchmark return and costs must the trade clear?
- Is success defined by a strict or non-strict target?

Without these choices, “probability of profit” is not well defined.

A Scheduled Entry and Exit



Net Present Value of a Long Position

Buy $n_0 > 0$ shares at price S_0 at time 0, then sell all shares at price S_T at time $T = N\Delta t$. For annual continuously compounded benchmark growth rate g_b, the frictionless time-0 NPV is:

$$\underbrace{\mathrm{NPV}(g_b, T)}_{\text{time-0 dollars}} = \underbrace{-\,n_0 S_0}_{\text{entry now}} + \underbrace{n_0 S_T e^{-g_b T}}_{\text{discounted exit}}.$$

Divide by the initial share cost to obtain a dimensionless present-value return:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\mathrm{NPV}(g_b, T)}{n_0 S_0} = e^{-g_b T}\frac{S_T}{S_0} - 1.$$

A positive price change alone need not clear the benchmark, spread, and costs.

Short Holding Period

When $|g_b|T$ is small, $e^{-g_bT} \approx 1$. The scaled NPV is then approximately the fractional change in share price:

$$\underbrace{\rho_T}_{\text{scaled NPV}} \approx \underbrace{\frac{S_T}{S_0} - 1}_{\text{price return}}.$$

Under the binomial lattice, $K_N \sim \mathrm{Binomial}(N, p)$ and:

$$\frac{S_T}{S_0} = u^{K_N} d^{N-K_N}.$$

Therefore the $N + 1$ terminal nodes give the complete discrete distribution of the approximate return.

Longer Holding Period

For a longer horizon, or whenever $|g_b|T$ is not small, retain the benchmark discount factor. At terminal node $K_N = k$:

$$\underbrace{\rho_T(k)}_{\text{present-value return}} = \underbrace{e^{-g_b N\Delta t}}_{\text{discount}} \underbrace{u^k d^{N-k}}_{\text{price ratio}} - 1, \qquad k = 0, \dots, N.$$

The node probability remains:

$$\mathbb{P}(K_N = k) = \binom{N}{k} p^k (1-p)^{N-k}.$$

The benchmark changes the return attached to each node. It does not change the real-world node probabilities specified by p. The matching one-step gross benchmark return from L3b is $R_b = e^{g_b \Delta t}$.

Cumulative Probability of a Terminal Target

Let $\rho_\star > -1$ be a dimensionless target. We ask for the probability that the scheduled terminal return strictly exceeds this target:

$$\mathbb{P}(\rho_T(K_N) > \rho_\star).$$

Since $u > d > 0$, the return $\rho_T(k)$ increases with the up-move count k. Therefore, the target event is equivalent to:

$$K_N > \tau(\rho_\star)$$

for a real-valued threshold $\tau(\rho_\star)$. The probability is a binomial tail, not the probability of one terminal node.

Solve the Return Inequality for the Up Count

Begin with the strict target at a node containing k up moves:

$$e^{-g_b N\Delta t} u^k d^{N-k} - 1 > \rho_\star.$$

Move positive terms, take logarithms, and use $\log(u/d) > 0$:

$$k \log(u/d) > \log(1+\rho_\star) + g_b N\Delta t - N \log d.$$

Thus the real-valued threshold is:

$$\underbrace{\tau(\rho_\star)}_{\text{count threshold}} = \frac{\log(1+\rho_\star) + g_b N\Delta t - N \log d}{\log(u/d)}.$$

Strict Targets Need the Next Integer

The event is $K_N > \tau(\rho_\star)$. Because K_N is integer valued, the smallest successful up count is:

$$\underbrace{k_{\min}}_{\text{first successful count}} = \lfloor \tau(\rho_\star) \rfloor + 1.$$

With $K_N \sim \mathrm{Binomial}(N, p)$, the target probability is:

$$\boxed{\mathbb{P}(\rho_T > \rho_\star) = \sum_{k=\max(0,k_{\min})}^{N} \binom{N}{k} p^k (1-p)^{N-k}.}$$

A non-strict event $\rho_T \geq \rho_\star$ uses a different integer boundary when a node lies exactly on the target.

Threshold Edge Cases

Keep the true integer $k_{\min}$ instead of clamping it into the feasible range.

- If $k_{\min} \leq 0$, every feasible terminal node succeeds, so the probability is 1.
- If $k_{\min} > N$, no feasible terminal node succeeds, so the probability is 0.
- At fixed N, increasing g_b raises the threshold and cannot increase the tail probability.
- Changing the horizon changes both the threshold and the distribution of K_N. Recompute the full probability.

Implementation: for a strict tail, use the complementary binomial CDF at $k_{\min} - 1$ only after handling the two edge cases.

Terminal Rules and First-Passage Rules Are Different

The binomial tail concerns liquidation at the scheduled date T. A take-profit or stop-loss rule monitored at each step depends on the first boundary crossing and is path dependent.

- Two paths can end at the same terminal node but cross a boundary at different earlier times.
- Propagate only probability mass that has not already crossed a boundary, then record the absorbed mass at each first crossing.
- A stop order becomes a market order after its trigger and need not fill at the trigger price. A limit order may not fill.

Terminal node probabilities alone cannot evaluate a monitored exit rule.

N-Ary Lattice Models

Replace the two branch factors with m positive factors $f_1, \ldots, f_m$. Let branch j occur with probability $p_j \geq 0$, where $\sum_{j=1}^{m} p_j = 1$.

After t steps, a recombining node is indexed by a branch-count vector:

$$\mathbf{x} = (x_1, \ldots, x_m), \qquad x_j \in \mathbb{Z}_{\geq 0}, \qquad \sum_{j=1}^{m} x_j = t.$$

Order does not matter because the branch factors are constant and multiplication commutes. Paths with the same count vector share one node.

N-Ary Node Values

For count vector $\mathbf{x}$ at level t, the node price is:

$$\underbrace{S_t(\mathbf{x})}_{\text{node price}} = S_0 \prod_{j=1}^{m} f_j^{x_j}.$$

Independent branch draws give the multinomial node probability:

$$\underbrace{\mathbb{P}(\mathbf{X} = \mathbf{x})}_{\text{node probability}} = \frac{t!}{x_1! \cdots x_m!} \prod_{j=1}^{m} p_j^{x_j}.$$

Distinct count vectors can still produce the same price if the factors have additional multiplicative relationships.

More Branches Create More States

The number of nonnegative count vectors at level t is the stars-and-bars count:

$$\underbrace{L_t}_{\text{nodes at level } t} = \binom{t+m-1}{m-1}.$$

The total number of nodes through height h is:

$$\underbrace{\sum_{r=0}^{h} L_r}_{\text{all stored nodes}} = \binom{h+m}{m}.$$

More branches give a finer discrete one-step distribution, but also increase the state count. They do not automatically improve out-of-sample forecasts.

N-Ary Example: Discretize, Build, Check

The companion notebook uses empirical log growth rates for one firm.

1. Divide the one-step growth sample into three equal-width bins.
2. Estimate one representative factor and one frequency for each bin.
3. Build each recombining state from its three branch counts.
4. Compute its multinomial probability and verify that all node probabilities at the selected level sum to one.
5. Plot price and annualized multistep log growth against probability mass.

Interpretation: the histogram of node locations is not a probability distribution unless each node is weighted by its probability mass.

Optional Advanced Material

First-passage exit rules advanced/first-passage/

Propagate only probability mass for positions that remain open. Compute the probabilities of reaching the take-profit boundary first, reaching the stop-loss boundary first, or remaining open through the horizon.

Execution-aware probability of profit advanced/execution/

Buy at the ask, sell at the bid, and include fees and slippage. Derive the new terminal-price and up-count thresholds, then compare them with the frictionless rule.

Notebooks: lectures/week-4/L4a/advanced, also linked from the Optional Advanced Material section of the lecture notebook.

Summary

Today we converted a lattice forecast into a complete terminal target event, derived its exact probability, and generalized the state space.

Key takeaways

- Profit is a cash-flow event: entry, exit, horizon, benchmark, costs, and target strictness must be specified.
- Monotone terminal targets become count tails: solve for the integer boundary and handle thresholds outside the feasible range.
- The state depends on the rule: monitored exits need path information, while N-ary terminal states use multinomial branch counts.

Next, we take the continuous-time limit and introduce stochastic price models.

Today: use the lattice to define the event before computing its probability.